

Homework 2 Solutions

Based on Lecture 2: Functions, Graphs, and Transformations.

Exercise 1. Find the domain and range of $f(x) = \frac{\sqrt{x-1}}{x-3}$.

Domain. The square root requires $x-1 \geq 0$, i.e. $x \geq 1$, and the denominator requires $x \neq 3$. So the domain is

$$\boxed{[1, 3) \cup (3, \infty)}.$$

Range. Substitute $t = \sqrt{x-1} \geq 0$, so $x = t^2 + 1$ and $x - 3 = t^2 - 2$. Then

$$f = \frac{t}{t^2 - 2} =: h(t), \quad t \geq 0, \quad t \neq \sqrt{2}.$$

h is strictly decreasing on each of $[0, \sqrt{2})$ and $(\sqrt{2}, \infty)$. Take any t_1, t_2 with $0 \leq t_1 < t_2$ lying in the same one of these two intervals. Then

$$\begin{aligned} h(t_1) - h(t_2) &= \frac{t_1}{t_1^2 - 2} - \frac{t_2}{t_2^2 - 2} = \frac{t_1(t_2^2 - 2) - t_2(t_1^2 - 2)}{(t_1^2 - 2)(t_2^2 - 2)} \\ &= \frac{t_1 t_2^2 - t_2 t_1^2 - 2t_1 + 2t_2}{(t_1^2 - 2)(t_2^2 - 2)} = \frac{t_1 t_2(t_2 - t_1) + 2(t_2 - t_1)}{(t_1^2 - 2)(t_2^2 - 2)} \\ &= \frac{(t_2 - t_1)(t_1 t_2 + 2)}{(t_1^2 - 2)(t_2^2 - 2)}. \end{aligned}$$

The numerator is positive: $t_2 - t_1 > 0$ by assumption, and $t_1 t_2 + 2 > 0$ since $t_1, t_2 \geq 0$. For the denominator, both t_1, t_2 lie in the same interval, so either $t_1^2 - 2 < 0$ and $t_2^2 - 2 < 0$ (both less than $\sqrt{2}$), or $t_1^2 - 2 > 0$ and $t_2^2 - 2 > 0$ (both greater than $\sqrt{2}$); either way the product $(t_1^2 - 2)(t_2^2 - 2) > 0$. So $h(t_1) - h(t_2) > 0$, i.e. $h(t_1) > h(t_2)$, proving h is strictly decreasing on each branch.

- On $[0, \sqrt{2})$: $h(0) = 0$, and as $t \rightarrow \sqrt{2}^-$ the denominator $t^2 - 2 \rightarrow 0^-$ with numerator positive, so $h(t) \rightarrow -\infty$. Since h is an unbroken, strictly decreasing curve from 0 down to $-\infty$, it takes every value in $(-\infty, 0]$.
- On $(\sqrt{2}, \infty)$: as $t \rightarrow \sqrt{2}^+$ the denominator $\rightarrow 0^+$, so $h(t) \rightarrow +\infty$; as $t \rightarrow \infty$, $h(t) \rightarrow 0^+$ but never reaches 0. Since h is an unbroken, strictly decreasing curve from $+\infty$ toward (but not including) 0, it takes every value in $(0, \infty)$.

The two branches together cover $(-\infty, 0] \cup (0, \infty)$, so

$$\boxed{\text{range} = (-\infty, \infty)}.$$

(Every real number, positive, negative, or zero, is attained: for example $f(4) = 1$, $f(\frac{9}{4}) = \frac{1}{2}/(-\frac{3}{4}) = -\frac{2}{3}$, and pushing x close to 3 from either side sends $f(x)$ to $\pm\infty$.)

Exercise 2. True or false, with a reason:

- the range of $y = x^2$ is all real numbers
- the graph of $y = 1/x$ crosses the y -axis
- $y = \sqrt{x}$ is the top half of $y = x^2$ turned on its side
- $y = x^3$ has domain restricted to $x \geq 0$

- $|x|$ and x^2 have the same range
- **False.** Squares are never negative, so the range of $y = x^2$ is $[0, \infty)$, not all of $\mathbb{R}$.
- **False.** The y -axis is $x = 0$, which is excluded from the domain of $y = 1/x$, so the graph never meets it.
- **True.** Turning the sideways parabola $x = y^2$ (equivalently $y^2 = x$) on its side and keeping only the upper half $y \geq 0$ gives exactly $y = \sqrt{x}$, since solving $x = y^2$ for $y \geq 0$ yields $y = \sqrt{x}$.
- **False.** $y = x^3$ is defined, and one-to-one, for every real x ; its domain is $(-\infty, \infty)$.
- **True.** Both $|x|$ and x^2 take every nonnegative value and no negative value, so both have range $[0, \infty)$.

Exercise 3. Decide whether each is a function of x , and whether it is even, odd, or neither:

- $y = 3x - 2$
- $x = y^2 + 1$
- $y = x^4$
- $y = x^3 + x^2$
- $y = 3x - 2$: this is a function of x (one output per input). Let $f(x) = 3x - 2$. Then $f(-x) = -3x - 2$. This equals neither $f(x) = 3x - 2$ nor $-f(x) = -3x + 2$, so f is **neither** even nor odd.
- $x = y^2 + 1$: solving for y gives $y = \pm\sqrt{x-1}$, two values of y for each $x > 1$, so this is **not a function of x** .
- $y = x^4$: this is a function of x . Let $f(x) = x^4$. Then $f(-x) = (-x)^4 = x^4 = f(x)$, so f is **even**.
- $y = x^3 + x^2$: this is a function of x . Let $f(x) = x^3 + x^2$. Then $f(-x) = -x^3 + x^2$. This is not equal to $f(x) = x^3 + x^2$ (so not even), and not equal to $-f(x) = -x^3 - x^2$ (so not odd, since the x^2 terms have opposite signs). Thus f is **neither**.

Exercise 4 (harder). Find the domain and range of $g(x) = \frac{1}{\sqrt{x^2 - 4}}$.

Domain. The expression under the square root must be positive (it cannot be zero, since it sits in a denominator), so $x^2 - 4 > 0$, i.e. $x^2 > 4$. Thus

$$\boxed{(-\infty, -2) \cup (2, \infty)}.$$

Range. Let $u = x^2 - 4$. As x ranges over the domain, u ranges over $(0, \infty)$: as $|x| \rightarrow 2^+$, $u \rightarrow 0^+$, and as $|x| \rightarrow \infty$, $u \rightarrow \infty$, with u taking every value in between by continuity. Then $\sqrt{u}$ ranges over $(0, \infty)$ as well, and so does its reciprocal $1/\sqrt{u}$ (the reciprocal of a positive number can be made arbitrarily large by taking u near 0, or arbitrarily close to 0 by taking u large, and every intermediate positive value is attained by continuity). Hence

$$\boxed{(0, \infty)}.$$

Exercise 5. Describe the transformations from the parent function in each, then sketch the resulting graph:

- $y = -3|x + 2| + 1$
- $y = \sqrt{x - 4} - 2$
- $y = \frac{2}{x + 1} + 3$

(a) $y = -3|x + 2| + 1$. Starting from $y = |x|$: shift left 2 (giving $|x + 2|$), scale vertically by 3 (steeper slopes, ± 3), reflect across the x -axis (the negative sign, so the graph opens downward), then shift up 1. The vertex moves from $(0, 0)$ to $(-2, 1)$.

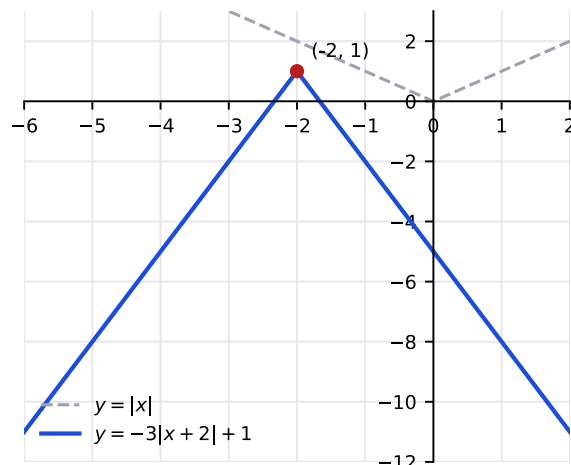


Figure 1: y equals $-3|x+2|+1$, obtained from the parent graph $y=|x|$ by a left shift, vertical scale, reflection, and upward shift, with vertex at $(-2,1)$

(b) $y = \sqrt{x-4} - 2$. Starting from $y = \sqrt{x}$: shift right 4 (giving $\sqrt{x-4}$, domain $x \geq 4$), then shift down 2. The starting point moves from $(0,0)$ to $(4,-2)$.

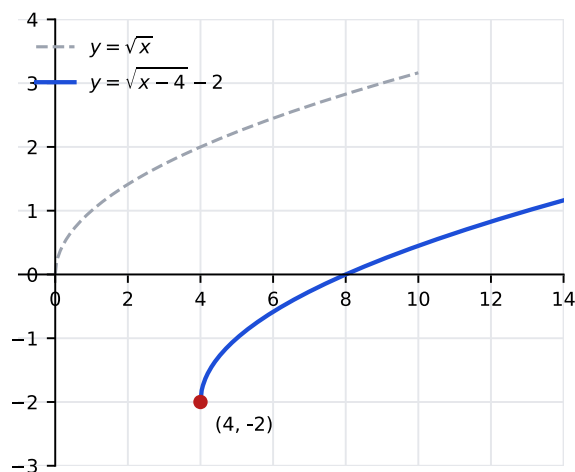


Figure 2: y equals square root of $(x-4)$ minus 2, obtained from the parent graph $y=\text{square root of } x$ by a right shift and a downward shift, starting point at $(4,-2)$

(c) $y = \frac{2}{x+1} + 3$. Starting from $y = 1/x$: shift left 1 (vertical asymptote moves from $x = 0$ to $x = -1$), scale vertically by 2, then shift up 3 (horizontal asymptote moves from $y = 0$ to $y = 3$).

Exercise 6. If $(2, -1)$ is on the graph of f , find the corresponding point on $y = f(x) + 4$, on $y = f(x - 3)$, and on $y = -f(-x)$.

Since $(2, -1)$ is on $y = f(x)$, we have $f(2) = -1$.

- $y = f(x) + 4$: from the transformation table, $f(x) + k$ sends (a, b) to $(a, b + k)$. With $a = 2, b = -1, k = 4$, the point is $\boxed{(2, 3)}$.
- $y = f(x - 3)$: $f(x - h)$ sends (a, b) to $(a + h, b)$. With $h = 3$, the point is $\boxed{(5, -1)}$.

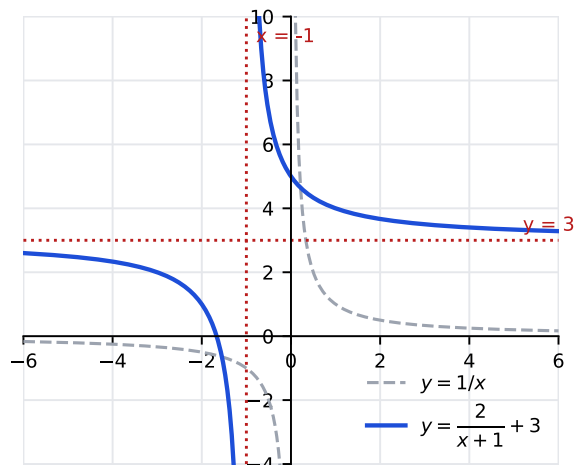


Figure 3: y equals 2 over $(x+1)$ plus 3, obtained from the parent graph $y=1/x$ by a left shift, vertical scale, and upward shift, with asymptotes $x=-1$ and $y=3$

- $y = -f(-x)$: reflect across the y -axis first ($f(-x)$ sends (a, b) to $(-a, b)$), then reflect across the x -axis ($-f(x)$ sends (a, b) to $(a, -b)$). Applying both to $(2, -1)$: first to $(-2, -1)$, then to $(-2, 1)$. Directly: if $x = -2$ then $-x = 2$, $f(-x) = f(2) = -1$, so $-f(-x) = 1$. The point is $\boxed{(-2, 1)}$.

Exercise 7. Find the line through $(-1, 4)$ and $(3, -8)$.

The slope is

$$m = \frac{-8 - 4}{3 - (-1)} = \frac{-12}{4} = -3.$$

Using point-slope form through $(-1, 4)$,

$$y - 4 = -3(x + 1),$$

so

$$\boxed{y = -3x + 1}.$$

Exercise 8. Sketch $y = -2(x + 1)^2 + 3$, then state its vertex, axis, opening direction, and range.

Comparing with $a(x - h)^2 + k$: since $x + 1 = x - (-1)$, we read off $a = -2$, $h = -1$, $k = 3$. Therefore:

- Vertex: $\boxed{(-1, 3)}$
- Axis: $\boxed{x = -1}$
- Opening direction: since $a = -2 < 0$, the parabola opens $\boxed{\text{downward}}$
- Range: the vertex is a maximum, so $\boxed{(-\infty, 3]}$

Exercise 9. Rewrite $y = 3x^2 - 12x + 7$ in vertex form.

Here $a = 3$, $b = -12$, $c = 7$, so

$$h = -\frac{b}{2a} = -\frac{-12}{2(3)} = 2, \quad k = c - \frac{b^2}{4a} = 7 - \frac{(-12)^2}{4(3)} = 7 - \frac{144}{12} = 7 - 12 = -5.$$

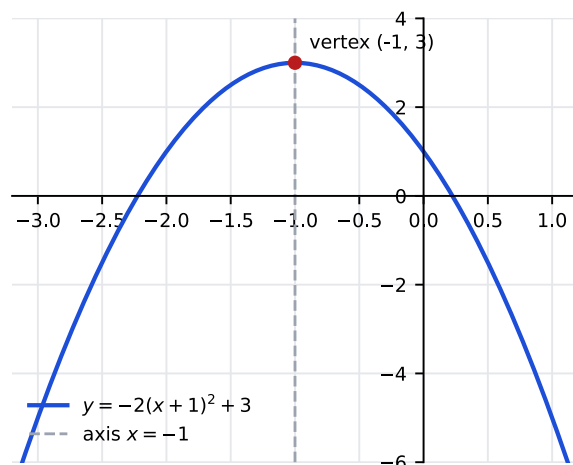


Figure 4: y equals $-2(x+1)$ squared plus 3, a downward parabola with vertex $(-1,3)$ and axis of symmetry $x=-1$

Therefore

$$y = 3(x - 2)^2 - 5.$$

Check by expanding: $3(x-2)^2 - 5 = 3(x^2 - 4x + 4) - 5 = 3x^2 - 12x + 12 - 5 = 3x^2 - 12x + 7$, which matches the original.

Exercise 10. Is $y = (x + 1)^3 - 4$ a transformation of $y = x^3$? Is $y = x^3 - 3x$? Explain.

$y = (x + 1)^3 - 4$: Yes. This has the form $a(x - h)^3 + k$ with $a = 1$, $h = -1$, $k = -4$ (since $x + 1 = x - (-1)$), which is $y = x^3$ shifted left 1 and down 4. As a shift of x^3 , it is strictly increasing everywhere and has no turning points.

$y = x^3 - 3x$: No. Factor it as $x^3 - 3x = x(x^2 - 3) = x(x - \sqrt{3})(x + \sqrt{3})$, which has three distinct real zeros: $-\sqrt{3}$, 0 , $\sqrt{3}$. A pure shift $a(x - h)^3 + k$ of $y = x^3$ is strictly monotonic (since x^3 is strictly increasing, or strictly decreasing if $a < 0$), so it can cross zero **at most once** — it cannot have three separate real zeros. Since $x^3 - 3x$ has three real zeros, it cannot be written in the form $a(x - h)^3 + k$, so it is not a transformation of $y = x^3$; between consecutive zeros the graph must turn around, giving it two turning points that no shift of x^3 can produce.

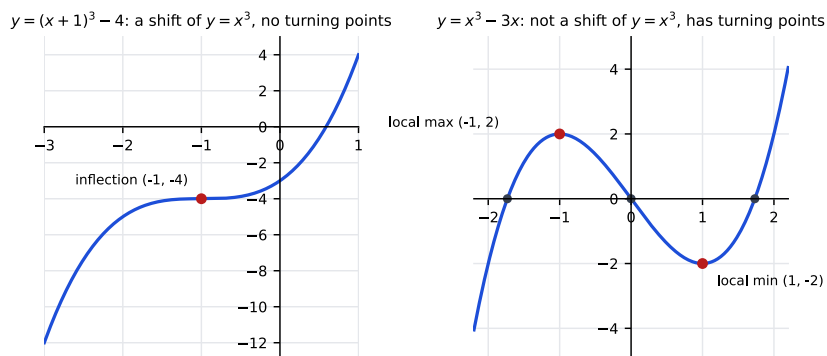


Figure 5: Cubic $(x+1)$ cubed minus 4, a shift of x cubed with no turning points, compared with x cubed minus $3x$, which is not a shift of x cubed and has a local max at $(-1, 2)$ and a local min at $(1, -2)$

Exercise 11. Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$$(f \circ g)(x) = f(g(x)) = f(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = \boxed{4x^2 - 12x + 10}.$$

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = 2(x^2 + 1) - 3 = 2x^2 + 2 - 3 = \boxed{2x^2 - 1}.$$

Exercise 12. Let $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$. Find the formula and domain of $(f \circ g)(x)$ and $(g \circ f)(x)$.

$(f \circ g)(x) = f(g(x)) = \sqrt{x^2 - 4}$. This requires x in the domain of g (all reals) with $g(x) = x^2 - 4 \geq 0$ in the domain of f , i.e. $x^2 \geq 4$. So

$$(f \circ g)(x) = \boxed{\sqrt{x^2 - 4}}, \quad \text{domain } \boxed{(-\infty, -2] \cup [2, \infty)}.$$

$(g \circ f)(x) = g(f(x)) = (\sqrt{x})^2 - 4$. This requires x in the domain of f , namely $x \geq 0$; once $x \geq 0$, $(\sqrt{x})^2 = x$ exactly, and g accepts any real input, so no further restriction applies. So

$$(g \circ f)(x) = \boxed{x - 4}, \quad \text{domain } \boxed{[0, \infty)}.$$

(Note that the formula $x - 4$ looks like it is defined for all x , but the domain of the composite is still restricted to $x \geq 0$, inherited from f .)

Exercise 13. For $y = 3(x - 2)^2 - 5$, state the intervals of increase and decrease and the minimum value.

Here $a = 3 > 0$, $h = 2$, $k = -5$, so the vertex $(2, -5)$ is a global minimum. The parabola decreases before the vertex and increases after it:

- Decreasing on $\boxed{(-\infty, 2]}$
- Increasing on $\boxed{[2, \infty)}$
- Minimum value $\boxed{-5}$ (attained at $x = 2$)

Exercise 14 (harder). The cubic $y = x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$ is an odd function with zeros at $-\sqrt{3}$, 0, and $\sqrt{3}$. Using only the end behavior and the zeros (no calculus): decide whether the graph is increasing or decreasing immediately to the left of $x = -\sqrt{3}$ and immediately to the right of $x = \sqrt{3}$; then explain why there must be a local maximum strictly between $x = -\sqrt{3}$ and $x = 0$, and (using the odd symmetry of the graph) a local minimum strictly between $x = 0$ and $x = \sqrt{3}$.

End behavior. For $y = x^3 - 3x$, the leading term x^3 dominates for large $|x|$, so as $x \rightarrow -\infty$, $y \rightarrow -\infty$, and as $x \rightarrow +\infty$, $y \rightarrow +\infty$.

A cubic has at most two turning points. A polynomial of degree 3 can change direction (from increasing to decreasing, or back) at most twice. We use this standard graphing fact without calculus.

Using up both turning points between the outer zeros. Between $-\sqrt{3}$ and 0 the function goes from 0 up to some value and back down to 0 (checking a test point, $x = -1$ gives $y = (-1)^3 - 3(-1) = 2 > 0$, so the graph is above the axis on this interval) — it must therefore turn around at least once inside $(-\sqrt{3}, 0)$. Likewise, between 0 and $\sqrt{3}$ the test point $x = 1$ gives $y = 1 - 3 = -2 < 0$, so the graph dips below the axis and must turn around at least once inside $(0, \sqrt{3})$. That already accounts for both of the cubic's turning points (one in each interval). Consequently, **no turning point remains available outside the interval $(-\sqrt{3}, \sqrt{3})$** , so the graph must be monotonic on $(-\infty, -\sqrt{3}]$ and on $[\sqrt{3}, \infty)$, matching the end behavior:

- Immediately to the left of $x = -\sqrt{3}$: the graph is $\boxed{\text{increasing}}$ (rising from $-\infty$ up to the zero at $-\sqrt{3}$, with no turning point available to interrupt it).
- Immediately to the right of $x = \sqrt{3}$: the graph is $\boxed{\text{increasing}}$ (continuing to rise toward $+\infty$, again with no turning point available).

Local maximum between $-\sqrt{3}$ and 0. The sign check above gives $y(-1) = 2 > 0$, while the endpoint values are $y(-\sqrt{3}) = y(0) = 0$. Thus the continuous graph rises above both endpoint values and must turn around at a **local maximum strictly between $x = -\sqrt{3}$ and $x = 0$.**

Local minimum between 0 and $\sqrt{3}$, by odd symmetry. Since $y = x^3 - 3x$ is odd, its graph is symmetric under $(x, y) \mapsto (-x, -y)$. The local maximum found at some point $(-c, d)$ with $-\sqrt{3} < -c < 0$ maps under this symmetry to the point $(c, -d)$ with $0 < c < \sqrt{3}$. A local maximum at $(-c, d)$ (graph increasing before, decreasing after) maps to a point where the graph is decreasing before and increasing after, i.e. exactly the definition of a **local minimum**, located at $(c, -d)$ with $0 < c < \sqrt{3}$ — strictly between $x = 0$ and $x = \sqrt{3}$, as required.

(Direct check: the actual turning points occur at $x = \pm 1$, which indeed satisfy $-\sqrt{3} < -1 < 0 < 1 < \sqrt{3}$; see the right-hand panel of the figure in Exercise 10, where the local max $(-1, 2)$ and local min $(1, -2)$ are marked.)

Exercise 15 (Challenge). Show that $f(x) = 1/x$ is decreasing on $(0, \infty)$ and decreasing on $(-\infty, 0)$, but give specific numbers $x_1 < x_2$ (with x_1 negative and x_2 positive) showing that f is *not* decreasing on its full domain.

Decreasing on $(0, \infty)$. Let $0 < x_1 < x_2$. Then

$$f(x_1) - f(x_2) = \frac{1}{x_1} - \frac{1}{x_2} = \frac{x_2 - x_1}{x_1 x_2}.$$

Since $x_2 > x_1$, the numerator $x_2 - x_1 > 0$; since $x_1, x_2 > 0$, the denominator $x_1 x_2 > 0$. So $f(x_1) - f(x_2) > 0$, i.e. $f(x_1) > f(x_2)$. This holds for every such pair, so f is decreasing on $(0, \infty)$.

Decreasing on $(-\infty, 0)$. Let $x_1 < x_2 < 0$. The same computation gives $f(x_1) - f(x_2) = \frac{x_2 - x_1}{x_1 x_2}$. Again $x_2 - x_1 > 0$, and now $x_1 x_2 > 0$ too (a product of two negative numbers). So again $f(x_1) - f(x_2) > 0$, i.e. $f(x_1) > f(x_2)$, so f is decreasing on $(-\infty, 0)$.

Not decreasing on the full domain. Take

$$\boxed{x_1 = -1, \quad x_2 = 1}.$$

Then $x_1 < x_2$, but

$$f(x_1) = \frac{1}{-1} = -1, \quad f(x_2) = \frac{1}{1} = 1,$$

so $f(x_1) = -1 < 1 = f(x_2)$. A decreasing function would require $f(x_1) > f(x_2)$, but here $f(x_1) < f(x_2)$ instead. Hence f is not decreasing across its full domain, even though it is decreasing on each of the two pieces separately.

Exercise 16. For each, decide whether $y = f(x)$ can be solved uniquely for x on the stated domain (that is, whether f is one-to-one there):

- $f(x) = 2x + 1$ on $\mathbb{R}$
- $g(x) = x^3$ on $\mathbb{R}$
- $g(x) = \frac{1}{x^2}$ on $\mathbb{R}$
- $f(x) = 2x + 1$: solving $y = 2x + 1$ gives $x = \frac{y-1}{2}$, a unique solution for every y . **One-to-one.**
- $g(x) = x^3$: x^3 is strictly increasing on all of $\mathbb{R}$ (it has no turning points), so distinct inputs give distinct outputs, and $y = x^3$ solves uniquely as $x = \sqrt[3]{y}$. **One-to-one.**

- $g(x) = \frac{1}{x^2}$: this is an even function, $g(-x) = g(x)$, so $g(1) = g(-1) = 1$ gives two different inputs with the same output. **Not one-to-one.**

Exercise 17. Find the inverse of $f(x) = \frac{x-1}{x+2}$ and state its domain and range.

The domain of f is $x \neq -2$. Write $y = \frac{x-1}{x+2}$, swap x and y , and solve for y :

$$x = \frac{y-1}{y+2}.$$

$$x(y+2) = y-1 \implies xy + 2x = y-1 \implies xy - y = -1 - 2x \implies y(x-1) = -1 - 2x.$$

$$y = \frac{-1-2x}{x-1} = \frac{2x+1}{1-x}.$$

So

$$\boxed{f^{-1}(x) = \frac{2x+1}{1-x}}.$$

The domain of f^{-1} is the range of f . Setting $\frac{x-1}{x+2} = 1$ gives $x-1 = x+2$, i.e. $-1 = 2$, which is impossible, so f never equals 1; every other value is attained (this matches f^{-1} 's own formula, which is undefined exactly at $x = 1$). Hence the range of f , and thus the domain of f^{-1} , is $\boxed{\{x : x \neq 1\}}$. The range of f^{-1} is the domain of f , namely $\boxed{\{y : y \neq -2\}}$.

Exercise 18 (Challenge). Let f be one-to-one and $g(x) = af(b(x-h)) + k$ with $a, b \neq 0$. Find g^{-1} in terms of f^{-1} .

Write $y = g(x) = af(b(x-h)) + k$ and solve for x step by step, undoing each operation in reverse order:

$$y - k = af(b(x-h)) \implies \frac{y-k}{a} = f(b(x-h)).$$

Since f is one-to-one, apply f^{-1} to both sides:

$$f^{-1}\left(\frac{y-k}{a}\right) = b(x-h).$$

$$\frac{1}{b} f^{-1}\left(\frac{y-k}{a}\right) = x-h \implies x = h + \frac{1}{b} f^{-1}\left(\frac{y-k}{a}\right).$$

Relabeling the input variable as x ,

$$\boxed{g^{-1}(x) = h + \frac{1}{b} f^{-1}\left(\frac{x-k}{a}\right)}.$$

Exercise 19 (Challenge). Prove directly from the definitions that $(f^{-1})^{-1} = f$.

Proof. Let $g = f^{-1}$. By the definition of inverse function, g has domain equal to the range of f , range equal to the domain of f , and satisfies

$$g(f(x)) = x \text{ for all } x \in \text{dom}(f), \quad f(g(y)) = y \text{ for all } y \in \text{dom}(g).$$

The two identities already say exactly that f reverses g : for every $y \in \text{dom}(g)$, $f(g(y)) = y$, and for every $x \in \text{dom}(f) = \text{range}(g)$, $g(f(x)) = x$. Thus f is an inverse function of g . An inverse, when it exists, is unique, so $g^{-1} = f$. Therefore

$$\boxed{(f^{-1})^{-1} = g^{-1} = f}. \quad \blacksquare$$

Exercise 20 (Challenge). Prove that if f is strictly increasing on an interval, then f is one-to-one there; and if f^{-1} exists, prove that f^{-1} is also strictly increasing.

Proof that f is one-to-one. Suppose f is strictly increasing on an interval I : for all $x_1, x_2 \in I$ with $x_1 < x_2$, $f(x_1) < f(x_2)$. Let $x_1, x_2 \in I$ with $x_1 \neq x_2$; relabel if necessary so that $x_1 < x_2$. By strict increase, $f(x_1) < f(x_2)$, so in particular $f(x_1) \neq f(x_2)$. Thus distinct inputs in I always give distinct outputs, which is exactly the definition of one-to-one. $\blacksquare$

Proof that f^{-1} is strictly increasing. Since f is one-to-one on I , f^{-1} exists on $J = f(I)$, the range of f restricted to I . Let $y_1, y_2 \in J$ with $y_1 < y_2$, and set $x_1 = f^{-1}(y_1)$, $x_2 = f^{-1}(y_2)$, so $x_1, x_2 \in I$, $f(x_1) = y_1$, $f(x_2) = y_2$. We must show $x_1 < x_2$.

Suppose, for contradiction, that $x_1 \geq x_2$.

- If $x_1 = x_2$, then $y_1 = f(x_1) = f(x_2) = y_2$, contradicting $y_1 < y_2$.
- If $x_1 > x_2$, then since f is strictly increasing and $x_2 < x_1$, we get $f(x_2) < f(x_1)$, i.e. $y_2 < y_1$, again contradicting $y_1 < y_2$.

Both cases are impossible, so $x_1 < x_2$, i.e. $f^{-1}(y_1) < f^{-1}(y_2)$. Since $y_1 < y_2$ were arbitrary elements of J , f^{-1} is strictly increasing on J . $\blacksquare$